

Rabiu Onoruoiza Mamman

Ph.D. Candidate, Mechanical Engineering | University of Iowa

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LinkedIn | Google Scholar

RESEARCH INTERESTS

Bio-inspired smart materials & adaptive structures · Artificial muscles & soft actuators · Multiscale mechanical characterization · Soft robotics & autonomous systems · Active flow control · Anti-fouling surface engineering

EDUCATION

Ph.D. in Mechanical Engineering

2022 – 2026 (Expected)

University of Iowa, Iowa City, IA, USA

- GPA: 4.0/4.0 | Fully funded: Dean's Excellence Fellowship & Graduate School Tuition Scholarship
- Dissertation: Bio-inspired smart materials, artificial muscles, and adaptive mechanical systems
- Advisor: Prof. Caterina Lamuta, Smart Multifunctional Material Systems Laboratory

M.S. in Mechanical Engineering

2022 – 2025

University of Iowa, Iowa City, IA, USA

- GPA: 4.0/4.0 | Concurrent with Ph.D. program

M.S. in Mechanical Engineering (Industrial & Production)

2016 – 2019

Federal University of Technology Minna, Nigeria

- GPA: 4.53/5.0
- Thesis: Analysis of offshore wind energy potential for power generation in Nigeria

B.Eng. in Mechanical Engineering

2007 – 2012

Ahmadu Bello University, Zaria, Nigeria

- Final-year project: Development and characterization of a bio-based hybrid polymer composite (polyester/glass fibre/orange peel particulates)

RESEARCH EXPERIENCE

Graduate Research Assistant

2024 – Present

Smart Multifunctional Material Systems Laboratory, University of Iowa, Iowa City, IA

Project 1: Development and Characterization of a Bistable Linear Actuator

- Designed and fabricated a vacuum-actuated bistable linear actuator that exploits snap-through instabilities to deliver impulsive actuation (~ 4.1 cm stroke in ~ 0.3 s) while lifting loads up to 0.5 kg at less than 65 kPa vacuum—enabling passive self-locking without continuous power input.
- Developed parametric CAD models and finite-element simulations (Abaqus) capturing negative-stiffness and bistable behavior in FDM PLA and Flexible 80A resin, establishing geometry–material design rules for reliable snap-through and stable latching.
- Engineered a precision pneumatic control platform integrating electro-pneumatic regulation, Arduino-based PWM-to-analog signal conditioning, laser displacement sensing (Keyence LK-G157), and NI-DAQ data acquisition for high-fidelity performance evaluation.
- Characterized peak actuation efficiency ($\leq 50.47\%$) and benchmarked stroke–load trade-offs across material configurations; results support manuscript accepted in *Smart Materials and Structures* (2026).

Graduate Research Assistant

2024 – 2025

*Smart Multifunctional Material Systems Laboratory, University of Iowa, Iowa City, IA***Project 2: Bio-inspired Active Vortex Generators for Underwater Flow Control**

- Developed artificial muscle-actuated deployable vortex generators for on-demand hydrodynamic flow control on underwater hydrofoils; published in *Robotics Reports* (2025).
- Experimentally demonstrated a 25–30% increase in lift coefficient and 8–20% drag reduction at high angles of attack and low Reynolds numbers—validated through flow visualization in a recirculating water channel.
- Implemented real-time electrothermal actuation with PID and L1 adaptive control driven by embedded thermistor feedback; integrated MATLAB/Simulink, ROS, and Arduino hardware into a unified test platform.

Graduate Research Assistant

2023 – 2024

*Smart Multifunctional Material Systems Laboratory, University of Iowa, Iowa City, IA***Project 3: Fouling Release Mechanism of an Octopus-Inspired Smart Skin**

- Invented and fabricated an octopus-inspired smart skin actuated by artificial muscles that achieves controlled out-of-plane surface deformation to actively release marine biofilms; published in *Advanced Functional Materials* (2024).
- Achieved $>90\%$ biofilm release efficiency at a low actuation voltage of ~ 3.2 V using real-time PID control, demonstrating low-energy anti-fouling capability.
- Developed and experimentally validated a physics-based predictive model linking biofilm critical strain to artificial muscle electrothermal actuation parameters.
- Characterized nanomechanical properties of biofilms and quantified release area using a Keyence laser confocal profilometer and optical microscopy.

Graduate Research Assistant

2024

*Smart Multifunctional Material Systems Laboratory, University of Iowa, Iowa City, IA***Project 4: Nanomechanical Properties and Wear Resistance of Palladium Diselenide (PdSe_2)**

- Performed nanoindentation and nanoscratch testing using a Hysitron TI-950 TriboIndenter to quantify hardness, Young's modulus, creep behavior, and strain-rate sensitivity of PdSe_2 for flexible electronics applications.

- Conducted strain-rate-dependent creep testing and extracted strain-rate sensitivity exponents relevant to long-term reliability of wearable and flexible electronic devices.
- Generated nanomechanical datasets characterizing ductility, elasticity, and wear resistance; contributed to a co-authored publication in *Materials Science and Engineering: B* (2024).

Graduate Research Assistant

2023 – 2024

*Smart Multifunctional Material Systems Laboratory, University of Iowa, Iowa City, IA***Project 5: Deployable Vortex Generators for UAVs Powered by Cephalopod-Inspired Artificial Muscles**

- Developed bio-inspired deployable vortex generators powered by twisted spiral artificial muscles (TSAMs) for adaptive aerodynamic flow control in unmanned aerial vehicles (UAVs); published in *iScience* (2023).
- Demonstrated stall delay and significant lift enhancement relative to clean airfoils and static VG baselines through low-Reynolds-number wind tunnel experiments.
- Integrated an electrothermal actuation model with L1 adaptive control for autonomous, closed-loop in-flight VG deployment.

Graduate Research Assistant

2016 – 2018

*Mechanical Engineering Laboratory, Federal University of Technology Minna, Nigeria***Project 6: Offshore Wind Energy Potential Analysis for Power Generation in Nigeria**

- Conducted offshore wind resource characterization and time-series analysis using probabilistic distribution models (Weibull, Rayleigh, exponential) across three coastal Nigerian sites.
- Evaluated site-specific wind speed characteristics, energy density, and capacity factors; matched wind profiles to commercial turbine models to quantify viable energy potential.
- Contributed to two peer-reviewed journal publications on offshore and onshore wind energy in Nigeria.

Undergraduate Research Assistant

2011 – 2012

*Materials Engineering Laboratory, Ahmadu Bello University, Zaria, Nigeria***Project 7: Bio-Based Hybrid Polymer Composite for Structural Applications**

- Developed and characterized a bio-based hybrid polymer composite using polyester resin reinforced with glass fibres and orange peel particulates.
- Established structure–property relationships for tensile, flexural, hardness, and fracture behavior; published findings in *Advanced Journal of Graduate Research* (2019).

PUBLICATIONS

† Denotes corresponding/lead author. 10 peer-reviewed publications | Full list on Google Scholar.

Journal Articles

- [1] Mamman[†], R. O., Garofalo, S., Bruno, L., Pagnotta, L., Risso, G., & Lamuta, C. (2026). Vacuum-driven artificial muscles exploiting snap-through instability for fast and efficient actuation. *Smart Materials and Structures*. Accepted manuscript. doi:10.1088/1361-665X/ae620b

- [2] **Mamman[†], R. O.**, Weerakkody, T.H., & Lamuta, C. (2025). Bioinspired active vortex generators for underwater flow control. *Robotics Reports*, **3**(1), 28350111251365627.
- [3] **Mamman[†], R. O.**, Johnson, T., Weerakkody, T.H., & Lamuta, C. (2024). Fouling release mechanism of an octopus-inspired smart skin. *Advanced Functional Materials*, **34**(41), 2406405. doi:10.1002/adfm.202406405
- [4] Uzun, U., Kotak, P., Shakib, M. A., **Mamman, R. O.**, Daws, S., Kuo, C.N., & Lue, C.S. (2024). Nanomechanical properties and wear resistance of palladium diselenide (PdSe₂) for flexible electronics. *Materials Science and Engineering: B*, **304**, 117357.
- [5] **Mamman[†], R. O.**, Kotak, P., Weerakkody, T.H., Johnson, T., Krebill, A., Buchholz, J., & Lamuta, C. (2023). Deployable vortex generators for low Reynolds number applications powered by cephalopod-inspired artificial muscles. *iScience*, **26**(12), 108369. doi:10.1016/j.isci.2023.108369
- [6] **Mamman[†], R. O.**, Adedipe, O., Lawal, S.A., Olugboji, O.A., & Nwachukwu, V.C. (2022). Analysis of offshore wind energy potential for power generation in three selected locations in Nigeria. *African Journal of Science, Technology, Innovation and Development*, **14**(3), 774–789.
- [7] Lawal, S.A., Mayaki, E.A., Adedipe, O., Okoro, U.G., & **Mamman, R. O.** (2020). Assessment of three selected locations in Nigeria as offshore windfarms using multi-criteria decision procedure. *ABUAD Journal of Engineering Research and Development*, **3**(1), 90–102.
- [8] **Mamman[†], R. O.**, & Ramalan, A.M. (2019). Mechanical and physical properties of polyester reinforced glass fibre/orange peel particulate hybrid composite. *Advanced Journal of Graduate Research*, **7**(1), 18–26. doi:10.21467/ajgr.7.1.18-26
- [9] Okafor, N.W., Adedipe, O., Usman, F.J., Jiya, J.Y., & **Mamman, R. O.** (2019). Production of biogas from chicken and goat wastes. *Nigerian Research Journal of Engineering and Environmental Sciences*.
- [10] Adedipe, O., Abolarin, M.S., & **Mamman, R. O.** (2018). A review of onshore and offshore wind energy potential in Nigeria. *IOP Conference Series: Materials Science and Engineering*, **413**(1), 012039. doi:10.1088/1757-899X/413/1/012039

TEACHING EXPERIENCE

Graduate Teaching Assistant — ME:6130 Novel Artificial Muscles and Sensors 2023–2025

Department of Mechanical Engineering, University of Iowa, Iowa City, IA

- Designed and delivered computer laboratory sessions in MATLAB and Simulink on mathematical modeling, simulation, and control of twisted, coiled, and spiraled artificial muscles.
- Instructed hands-on fabrication and characterization laboratories; guided students through manufacture and experimental testing of novel artificial muscle systems.
- Evaluated student performance through grading of examinations, problem sets, and laboratory reports; proctored assessments for a graduate-level course.

Graduate Teaching Assistant — ME:4120 Modern Engineering Materials for Mechanical Design 2023–2025

Department of Mechanical Engineering, University of Iowa, Iowa City, IA

- Developed and delivered senior undergraduate lecture content on modern engineering materials, including polymers, composites, and advanced functional materials.
- Taught applied Finite Element Analysis (FEA) for structural simulation and introduced constitutive modeling of viscoelastic and hyperelastic materials.
- Mentored undergraduate students during office hours; graded homework assignments and proctored examinations.

TALKS & PRESENTATIONS

- Vacuum-Driven Artificial Muscles Exploiting Snap-Through Instability for Fast and Efficient Actuation. *University of Iowa College of Engineering Research Open House*, Iowa City, IA, USA (2026).
- Bioinspired Active Vortex Generators for Enhanced Underwater Flow Control. *ASME Smart Materials, Adaptive Structures, and Intelligent Systems (SMASIS) Conference*, St. Louis, MO, USA (2025).
- Active Vortex Generator for Underwater Flow Control. **[Winner, Best Poster]** *University of Iowa College of Engineering Research Open House*, Iowa City, IA, USA (2025).
- Physics-Based Model for Biofilm Detachment. *ASME SMASIS Conference*, Atlanta, GA, USA (2024).
- Bio-inspired Vortex Generators for Stall Delay. *ASME SMASIS Conference*, Austin, TX, USA (2023).
- Active Flow Control Using Artificial Muscles. *University of Iowa College of Engineering Research Open House*, Iowa City, IA, USA (2023).

HONORS & AWARDS

Winner, Research Open House Best Poster Presentation	University of Iowa, 2025
Graduate Student of the Year — Golden Torch Award	National Society of Black Engineers, 2025
College of Engineering Ambassador Travel Grant	University of Iowa, 2024
Dean's Excellence Fellowship	University of Iowa, 2022
Full Graduate School Tuition Scholarship	University of Iowa, 2022
Best Project Award	Association of Commonwealth Universities, 2018
Summer School Bursary	ACU / Chinese University of Hong Kong, 2018

TECHNICAL SKILLS

Computational & Simulation: Abaqus (FEA), MATLAB/Simulink, CREO Parametric, AutoCAD (transferable to SolidWorks), Python, ImageJ, Power BI, Microsoft Office Suite

Control Systems: PID control, L1 adaptive control, Arduino (PWM & analog I/O), Robot Operating System (ROS)

Fabrication & Prototyping: SLA and FDM 3D printing, resin molding, spin coating, hand layup, thin-walled polymer fabrication

Instrumentation & Metrology: Hysitron TI-950 TriboIndenter (nanoindentation, nanoscratch, wear testing), TestResources tensile/DMA systems, Keyence LK-G157 laser displacement sensor, Keyence laser confocal profilometer, NI-DAQ data acquisition, pressure/vacuum regulation, sensor calibration and troubleshooting

Experimental Methods: Flow visualization, hydrodynamic and aerodynamic testing, optical microscopy, real-time electrothermal characterization, nanomechanical testing

SERVICE & LEADERSHIP

Laboratory & Safety Manager, Smart Multifunctional Material Systems Lab 2024 – Present
College of Engineering, University of Iowa, Iowa City, IA

- Oversee laboratory safety compliance, equipment maintenance schedules, and onboarding protocols for new graduate and undergraduate researchers.

Grant Review Committee Member, Graduate & Professional Students Government (GPSG) 2023 – Present
University of Iowa, Iowa City, IA

- Review and evaluate research funding proposals submitted by graduate students; advise on resource allocation decisions.

Apex Member, National Society of Black Engineers (NSBE) 2023 – Present
University of Iowa Chapter, Iowa City, IA

- Contribute to community outreach, professional development initiatives, and peer mentoring within the chapter.

REFERENCES

Prof. Caterina Lamuta

Robert and Virginia Wheeler Faculty Fellow in Engineering; Associate Professor, Mechanical Engineering
University of Iowa, Iowa City, IA, USA
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Dr. Thilina Weerakkody

Senior Lecturer in Mechanical Engineering
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